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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001
B. Sc DEGREE EXAMINATION APRIL 2021
BC304 - ANALYTICAL BIOCHEMISTRY - I

Time: 3 hrs **Max Marks: 75**

SECTION–A (25X1=25)
Answer **ALL** Questions

I. Choose the correct answer

1. The factor which affect electrophoretic mobility of the sample is
   a. Charge b. Size c. Shape d. All of the above

2. The anionic detergent used in electrophoresis is
   a. SDS b. Urea c. CTAB d. Iodoacetate

3. In isopycnic centrifugation, the separation of particle is based on
   a. Charge b. Size c. Density d. None

4. In affinity chromatography, the common ligand for ribosomal RNA is
   a. Heparin b. Tryptophan c. Lysine d. poly U

5. SDS give _____ charge to the denatured polypeptides
   a) Postive b) Negative c) Neutral d) None

6. _____ is a measure of the acidity or basicity of an aqueous solution.
   a) pH b) pKa c) pOH d) PPM

7. _____ makes use of high centrifugal force for studying properties of biological particles.
   a) Ultracentrifugation b) Desktop c) Clinical d) Preparative

8. One Svedberg unit is

a)10^{-11} sec b)10^{-12} sec c)10^{-13} sec d)10^{-14} sec

9. The most common absorbent used in TLC is

a) cellulose acetate b) silica gel c) kieselguhr G d) none

10.Organelles are separated by

a) Spectroscopy b) centrifugation

c) electrophoresis d) Chromatography

II. True or false

11. SDS is a cationic detergent.
12. One svedberg unit is 1 x 10^{-13} sec.
13. Rotors regulates temperature in cooling centrifuges.
14. Buffers cannot resist change in pH.
15. Proteins cannot be separated in cooling centrifuges.

Answer in one or two sentences

16. Define pH.
17. Define Rf.
18. What is RCF?
19. List out any two factors that influence sedimentation rate.
20. Expand SDS-PAGE.
21. List few adsorbents of TLC.
22. What is centrifugal force?
23. Give any one application of TLC
24. Define buffer
25. What is rotor?

SECTION – B (4X5=20)

Answer any **FOUR** Questions

26. Mention few applications of buffers.
27. Explain the technique of isopycnic centrifugation.
28. Bring out the procedure and applications of affinity chromatography.
29. Derive Henderson-Hasselbach equation?
30. Write a note on paper chromatography
31. Write a short note on density gradient centrifugation.

SECTION –C (3X10=30)

Answer any **THREE** Questions

32. How is SDS-PAGE done for the determination of molecular weight of proteins?
33. Describe the principle and applications of glass electrode.
34. Explain in detail the principle and procedure of GLC.
35. How to determine the molecular weight of proteins by analytical ultracentrifuge?
36. Describe in detail the principle, procedure and applications of HPLC.

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